

Similarities and Differences Across Bipolar Disorder Subtypes Among Adolescents

Emma Green, BSc (Hons),¹ Simina Toma, MD, MSc,² Jordan Collins, PhD,² Lisa Fiksenbaum, PhD,²
Vanessa Timmins, MSW,² Danielle Omrin, MSW,² and Benjamin I. Goldstein, MD, PhD²

Abstract

Objectives: To compare demographic, clinical, and familial characteristics across bipolar disorder (BD) subtypes in adolescents.

Methods: A total of 168 participants, 13 to 19 years of age, with BD-I ($n=41$), BD-II ($n=68$), or operationalized BD—not otherwise specified (NOS) ($n=59$) were recruited from a tertiary subspecialty clinic at an academic health sciences center. Diagnoses were determined using the semistructured K-SADS-PL (Schedule for Affective Disorders and Schizophrenia for School-Age Children—Present and Lifetime Version) interview. Omnibus analyses were followed up with *post hoc* pairwise comparisons.

Results: After controlling for age, race, and living with both natural parents, BD-I was associated with greater functional impairment, increased rates of psychiatric hospitalization, psychosis, and lifetime exposure to second-generation antipsychotics and lithium, less self-injurious behavior, less anxiety disorders, and less severe worst lifetime depression and lower levels of emotional dysregulation and lability compared with both BD-II and BD-NOS. Lifetime most severe manic symptoms were highest in BD-I, lowest in BD-NOS, with BD-II intermediate. Lifetime exposure to psychosocial treatment followed the opposite pattern: lowest in BD-I, highest in BD-NOS, with BD-II intermediate. Variables for which there were no significant between-group differences included suicidal ideation, suicide attempts, comorbidities other than anxiety, or family history of BD.

Conclusion: Among observed differences, most distinguish BD-I from other subtypes, whereas few variables differed between BD-II and BD-NOS. Different BD subtypes share important similarities in multiple clinical and familial characteristics, including family history of BD. Present findings support and extend knowledge regarding the course and outcome of bipolar youth study operationalized definition of BD-NOS. Further research is warranted to evaluate intermediate phenotypes and treatment strategies that address these subtype-related differences.

Keywords: bipolar disorders, adolescents, subtypes

Introduction

BIPOLAR DISORDER (BD) is a recurrent mood disorder that is associated with significant functional impairment and symptomatic burden (Judd et al. 2008; Freeman et al. 2009). This is especially true when onset occurs in childhood or adolescence (Goldstein and Birmaher 2012). Childhood and adolescent-onset BD is associated with a more severe, adverse, and complicated course of the disorder, as reflected by greater time spent in mixed episodes, greater proportion of time spent symptomatic, and more frequent polarity switches (Birmaher and Axelson 2006). BD-I is characterized by the presence of at least one clear lifetime manic or

mixed episode, most often but not necessarily accompanied by depressive episodes. BD-II is characterized by the presence of at least one lifetime hypomanic episode and at least one lifetime major depressive episode (American Psychiatric Association 2013). In recent years, it has become increasingly recognized that BD occurs across a spectrum, and that bipolar-spectrum presentations other than BD-I and BD-II are also highly impairing and share many similarities with BD-I and BD-II (Axelson et al. 2006; Merikangas et al. 2011).

BD—not otherwise specified (BD-NOS) is a subtype of BD that lacks the clear operationalized definitions of BD-I or II (Fristad and Algorta 2013). For this reason, the course and outcome of bipolar

¹Department of Psychology, University of Waterloo, Waterloo, Canada.

²Department of Psychiatry, Centre for Youth Bipolar Disorder, Sunnybrook Health Sciences Centre, Toronto, Canada.

Funding: No funding was received for this article.

youth (COBY) study operationalized BD-NOS as comprising distinct period(s) of abnormally elevated, expansive, or irritable mood plus: (1) at least two manic symptoms (three if mood is irritable only) that are clearly associated with the onset of affective symptoms; (2) clear change in functioning associated with the onset of affective symptoms; (3) presence of the elated and/or irritable mood and manic symptoms for a significant part of the day (minimum of 4 hours, although does not have to be expressed consecutively); and (4) minimum of 4 days (not necessarily consecutive) meeting criteria 1–3 over the person's lifetime (Birmaher et al. 2006). In the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), the term BD-NOS has been replaced by the term, 'other specified bipolar and related disorder' (American Psychiatric Association 2013). Although the terms are essentially interchangeable, this article uses the term BD-NOS, as we used the COBY BD-NOS operationalized criteria, and for consistency with most previous studies. This study has the additional benefit of providing support for the COBY definition of BD-NOS, by its use as a criteria variable.

Studies of adults have examined the rates of BD and its subtypes on a worldwide scale; subthreshold BD (a proxy for NOS) has a lifetime prevalence of 1.4%, greater than that of BD-I (0.6%) or BD-II (0.4%) (Merikangas et al. 2011). BD-I has been associated with higher rates of psychiatric hospitalization and suicide attempts (Parker et al. 2013), whereas BD-II is associated with more depressive episodes (Judd et al. 2003). An international large-scale epidemiologic study found that subthreshold BD, like BD-I and BD-II, is associated with social and occupational impairment and high rates of comorbidities (Merikangas et al. 2011).

Several studies have compared differences across BD subtypes among youth specifically. As in adults, BD-I is associated with higher rates of elated mood, psychiatric hospitalization, and functional impairment (Birmaher et al. 2006). BD-II, by comparison, is associated with greater burden of depression symptoms and higher rates of anxiety disorders (Masi et al. 2007). Both these subtypes have been shown to have faster rates of recovery with increased chances of relapse/recurrence compared with BD-NOS. Accordingly, BD-NOS is associated with more persistent symptoms, slower recovery, and less euthymic periods (Birmaher et al. 2009). BD-NOS is associated with greater functional impairment compared with BD-II, despite demonstrating lower scores on cross-sectional measures of mania and depression. Furthermore, it is associated with an earlier age of onset when compared with both BD-I and -II (Birmaher et al. 2006; Masi et al. 2007). As in adults, the BD subtypes do demonstrate similarities, including rates of suicidality, comorbid substance use disorders (SUDs), and elevated family history of mood disorders (Goldstein et al. 2005, 2013). In particular, youth with BD have been shown to have increased risk of familial BD in first-degree relatives (as compared with controls and youth with attention-deficit/hyperactivity disorder [ADHD]). Youth with BD were also at increased risk of familial major depressive disorder in first-degree relatives (compared with youth with ADHD) (Wozniak et al. 2017).

Given the paucity of literature comparing BD subtypes in adolescents, we set out to examine this topic in a relatively large sample of Canadian adolescents with BD. Based on the available literature, we expected that the similarities across subtypes would exceed the differences, and that BD-NOS would be characterized by functional impairment, comorbidity, and risk indicators that support the validity of the COBY study's BD-NOS operationalization.

Methods

Sample and setting

This study included 168 participants, 13–19 years of age, with a diagnosis of BD-I ($n = 41$), BD-II ($n = 68$), or BD-NOS ($n = 59$). Participants were recruited from a tertiary subspecialty clinic in an academic health sciences center in Toronto, Canada. All participants and their parents/guardians provided written informed consent after discussing the study with a member of the research staff. The study was approved by the local Research Ethics Board.

Participant assessment

Diagnoses (current and lifetime) were assessed using the Schedule for Affective Disorders and Schizophrenia for School-Age Children—Present and Lifetime Version (K-SADS-PL) (Kaufman et al. 1997). The K-SADS Mania Rating Scale (MRS) and the depression section of the K-SADS-P (DEP-P) were used in place of the K-SADS-PL mood sections (Axelson et al. 2003). All interviewers had a university degree in psychology or a related mental health field and had completed comprehensive K-SADS-PL training under the supervision of the senior author, a licensed child-adolescent psychiatrist (B.I.G.). Additional information about past and current psychosocial treatment was also systematically ascertained during the K-SADS-PL interview.

Age of onset was defined as the age at which a participant met DSM-IV criteria for a manic, hypomanic, or depressive episode. For participants with a diagnosis of BD-NOS, age of onset was defined as the age that the participant met the COBY criteria for BD-NOS (Axelson et al. 2006). General levels of functioning were evaluated using the Children's Global Assessment Scale (C-GAS), for both current and most impaired period of participant's life (Shaffer et al. 1983). Socioeconomic status was evaluated using the Hollingshead Four Factor Index of Socioeconomic Status (Hollingshead 1975). History of lifetime physical and/or sexual abuse was ascertained using the K-SADS-PL Post-Traumatic Stress Disorder Screen (Kaufman et al. 1997). Family psychiatric history of first- and second-degree relatives was ascertained using the Family History Screen (Weissman et al. 2000). Mood lability was ascertained using self-report and parent-report versions of the Children's Affective Lability Scale (CALS) (Gerson et al. 1996). Emotional dysregulation and borderline personality-spectrum symptoms were evaluated using the Life Problems Inventory (LPI) (Rathus et al. 2015).

Data analysis

The three BD subtype groups were compared using Pearson's chi-square or Fisher's exact test for categorical variables and analysis of variance (ANOVA) or Kruskal–Wallis tests for continuous variables, as appropriate. After the initial analyses, all variables for which there were between-group differences, using a liberal selection criteria of $p < 0.2$, underwent a second series of multivariable analyses, controlling for age, race, and living with both natural parents. Analyses of covariance (ANCOVAs) were performed for continuous variables and multiple logistic regression models were built for dichotomous variables, alongside pairwise *post hoc* tests (Table 2). Table 2 provides descriptive statistics for all variables considered for this study. If ANOVA for a variable was not associated ($p < 0.2$) with BD subtype, multivariable analysis was not conducted for that variable. As indicated in Table 2, those variables for which ANOVA was not significant list “—” under the

related tests statistics and p -value, as ANCOVA was not undertaken for those variables. IBM Statistical Package for the Social Sciences (SPSS) 24 Statistics software was used to perform statistical analyses (IBM Corp 2016).

Results

Demographic information for the three BD subtypes is provided in Table 1. There were significant differences between subtypes on the demographic factors of age, race, and living with both natural parents, which were therefore included as covariates in multivariable analyses. There were no significant differences between socioeconomic status and history of abuse between the subtypes.

Clinical information for the three BD subtypes is given in Table 2.

Results demonstrate that participants with BD-I differed significantly from those with BD-II and BD-NOS in 12 of 16 variables. Those with BD-I had the highest rates of psychiatric hospitalization ($p < 0.001$), psychosis ($p < 0.001$), and previous treatment with second-generation antipsychotics (SGAs; $p = 0.002$) and lithium ($p < 0.001$). In contrast, BD-I participants had the lowest rates of self-injurious behavior ($p = 0.001$) and comorbid diagnosis of anxiety ($p = 0.005$), the lowest ratings of both emotional dysregulation/borderline personality spectrum symptoms ($p = 0.009$) and mood lability ($p = 0.001$), the lowest rates of previous psychosocial treatment ($p = 0.018$), and were least likely to have a family history of depression ($p = 0.002$) or anxiety ($p = 0.005$).

The three subtypes differed significantly from one another for two variables. Most severe lifetime manic symptoms were highest in BD, lowest in BD-NOS, and intermediate in BD-II ($p < 0.001$). The opposite pattern was observed for previous psychosocial treatment, which was most common in BD-NOS, least common in BD-I, with BD-II intermediate ($p = 0.018$).

Participants with BD-II significantly differed from those with BD-I and BD-NOS for two variables. For past symptoms of depression, participants with BD-II had the most severe presentation compared with participants with BD-I ($p < 0.001$); however, participants with BD-NOS did not differ from those diagnosed with

BD-II. In terms of age of onset, participants with BD-NOS had the youngest age of onset ($p < 0.001$), in comparison with the other two subtypes.

Discussion

This study compared demographic, clinical, and familial characteristics among adolescents with BD-I, BD-II, and BD-NOS. Although between-group differences were observed for several variables, the majority of variables did not differ across BD subtypes. Given the ongoing questions, both in the adult literature and the youth literature (Towbin et al. 2013), as to whether parsing non-BD-I subtypes is warranted, present findings contribute additional data to inform this discussion in youth. Ultimately, present findings provide further support for the perspective that BD subtypes in youth are sufficiently similar so as to support the concept of a bipolar spectrum while maintaining sufficient differences to support the distinction between these subtypes. Present findings also support the conclusion that differences across BD subtypes in this sample were mostly observed when contrasting BD-I from the other BD subtypes. Second, the findings of this study are a source of support for the COBY-based definition of BD-NOS, as BD-NOS did demonstrate differences from the other subtypes of this study. Before turning to the implications of this overall pattern of findings, we first contextualize the individual findings within the existing literature.

The overall key finding of this study was that subtype differences among adolescents with BD are primarily related to differences between BD-I and other subtypes. Higher rates of psychosis and psychiatric hospitalization in BD-I youth than the other subtypes align with previous findings from clinical and epidemiologic samples (Birmaher and Axelson 2006; Birmaher et al. 2009; Van Meter et al. 2016). Contrasting these expected, and arguably self-evident, differences, the present findings replicate previous findings that are perhaps less expected, such as the higher rates of psychiatric hospitalization among youth with BD-NOS compared with BD-II (Rizzo et al. 2007). Inpatient treatment has been previously associated with greater global impairment (Rizzo et al. 2007). This is consistent with the findings of this study as

TABLE 1. DEMOGRAPHIC CHARACTERISTICS OF ADOLESCENTS WITH BIPOLAR DISORDER

	Overall (N = 168)		BD-I (n = 41)		BD-II (n = 68)		BD-NOS (n = 59)		χ^2/F value	p	Post hoc ^a	Effect size (η_p^2 or Cramer's V) ^b
	Mean	SD	Mean	SD	Mean	SD	Mean	SD				
Age	16.26	1.48	17.07	1.19	16.00	1.56	15.98	1.40	9.09	<0.001	AB	0.10
Socioeconomic status	49.28	12.89	51.58	12.87	48.48	13.10	48.64	12.68	0.835	0.436	—	0.01
	n	%	n	%	n	%	n	%				
Race (white)	137	81.1	28	68.29	58	85.29	51	86.44	5.755	0.056	—	0.19
Sex (male)	63	37.50	18	43.90	19	27.94	26	44.07	4.454	0.108	—	0.16
Living with both natural parents	93	55.00	29	70.73	39	57.35	25	42.37	8.563	0.014	B	0.23
Past history of abuse	23	13.70	6	14.63	10	14.71	7	11.86	0.325	0.850	—	0.04

A, BD-I versus BD-II; B, BD-I versus BD-NOS.

^aPost hoc presented for ANCOVA; —, nonsignificant in ANOVA analyses.

^bPartial eta-squared reported for ANCOVA and Nagelkerke R -squared for logistic regressions.

ANCOVA, analysis of covariance; ANOVA, analysis of variance; BD, bipolar disorder; BD-NOS, BD—not otherwise specified; SD, standard deviation.

TABLE 2. ANALYSES OF COVARIANCE AND *POST HOC* ANALYSES OF CLINICAL CHARACTERISTICS, COMORBIDITIES, TREATMENT HISTORY, DIMENSIONAL SCALES, AND FAMILY PSYCHIATRIC HISTORY

	Overall (N=168)		BD-I (n=41)		BD-II (n=68)		BD-NOS (n=59)		Wald/F value	p	Post hoc ^a	Effect size ^b
	Mean	SD	Mean	SD	Mean	SD	Mean	SD				
Clinical characteristics												
Age-of-onset BD	14.50	2.74	16.01	2.21	14.60	2.08	13.29	3.21	8.725	<0.001	BC	0.11
C-GAS (past)	41.98	8.66	36.44	10.15	43.59	6.12	43.97	8.54	9.468	<0.001	AB	0.11
C-GAS (current episode)	53.27	10.43	53.66	13.11	53.91	9.25	52.28	9.71	—	—	—	
MRS (past)	28.68	9.65	36.12	9.48	28.44	7.48	23.78	8.82	20.570	<0.001	ABC	0.20
MRS (current)	18.12	12.19	15.56	16.38	20.68	10.47	16.98	10.16	—	—	—	
Dep-P (past)	31.26	10.43	26.98	12.94	34.56	7.85	30.43	10.00	7.612	0.001	A	0.09
Dep-P (current)	19.21	13.08	10.63	11.17	22.57	12.89	21.31	12.05	13.036	<0.001	AB	0.14
	n	%	n	%	n	%	n	%				
Psychiatric hospitalization	82	49.10	36	87.80	24	35.29	22	37.29	21.269	<0.001	AB	0.31
Self-injurious behavior	92	54.40	9	21.95	43	63.24	40	67.80	14.222	0.001	AB	0.23
Suicidal ideation	104	61.50	19	46.34	46	67.65	39	66.10	1.422	0.491		0.11
Suicide attempt	40	23.70	8	19.51	19	27.94	13	22.03	—	—	—	0.02
Psychosis	20	11.80	15	36.60	2	2.90	3	5.00	17.093	<0.001	AB	0.30
Comorbid diagnoses												
ADHD	72	42.86	14	34.15	31	45.59	27	45.76	—	—	—	0.03
SUD	53	31.7	11	26.82	24	35.29	18	30.51	—	—	—	0.05
CD	12	7.1	2	4.88	4	5.88	6	10.17	—	—	—	0.03
ODD	55	32.74	7	17.07	21	30.88	27	45.76	5.707	0.058		0.10
Any anxiety	132	78.10	25	60.98	58	85.29	49	83.05	10.473	0.005	AB	0.17
Treatment												
Antimaniac/anticonvulsants	17	10.10	4	9.80	6	8.80	7	11.7	.737	0.692		0.04
Second-generation antipsychotic	90	53.30	34	82.93	30	44.12	26	44.07	12.664	0.002	AB	0.19
Lithium	29	17.20	19	46.34	3	4.41	7	11.86	18.369	<0.001	AB	0.32
Lamotrigine	8	4.7	1	2.44	3	4.41	4	6.78	—	—	—	0.08
Non-SSRI antidepressants	18	10.70	1	2.40	9	13.20	8	13.30	4.554	0.103		0.12
Stimulants	36	21.3	7	17.07	14	20.59	15	25.42	—	—	—	0.03
SSRI antidepressants	63	37.30	10	24.4	27	39.70	26	43.30	3.395	0.183		0.08
Any psychosocial treatment	133	81.10	26	63.4	60	89.6	47	83.9	8.737	0.013	AB	0.14
	Mean	SD	Mean	SD	Mean	SD	Mean	SD				
Dimensional scales												
LPI total score	32.60	14.05	25.63	11.21	33.31	10.70	36.11	17.45	4.864	0.009	AB	0.06
Total CALSC	30.96	17.81	19.95	19.41	34.03	16.41	35.05	15.17	7.511	0.001	AB	0.09
Total CALSP	23.58	15.79	18.91	18.12	25.32	15.63	24.56	14.11	.514	0.599		0.01
	n	%	n	%	n	%	n	%				
Family psychiatric history (first or second degree)												
SUD	80	47.30	13	31.70	35	51.50	32	53.30	2.050	0.359		0.08
Depression	131	77.50	22	53.66	62	91.18	47	79.66	12.979	0.002	AB	0.20
CD	66	39.1	13	31.71	28	41.18	25	42.37	—	—	—	0.04
Suicide attempt	46	27.50	7	17.50	20	29.90	19	31.70	—	—	—	0.04
Mania/hypomania	81	47.90	15	36.59	40	58.82	26	44.07	3.303	0.192		0.11
ADHD	57	33.70	7	17.07	31	45.59	19	32.20	5.757	0.056		0.14
Anxiety	100	59.20	15	36.59	47	69.12	38	64.41	10.777	0.005	AB	0.14

A, BD-I versus BD-II; B, BD-I versus BD-NOS; C, BD-II versus BD-NOS.

^a*Post hoc* presented for ANCOVA; —, nonsignificant in ANOVA.

^bPartial eta-squared reported for ANCOVA and Nagelkerke *R*-squared for logistic regressions.

ADHD, attention-deficit/hyperactivity disorder; ANCOVA, analysis of covariance; ANOVA, analysis of variance; BD, bipolar disorder; BD-NOS, BD—not otherwise specified; C-GAS, Children's Global Assessment Scale; CALSC, Children's Affective Liability Scale (child report); CALSP, Children's Affective Liability Scale (parent report); CD, conduct disorder; Dep-p, depression section of the K-SADS-P; LPI, Life Problems Inventory; MRS, Mania Rating Scale; ODD, oppositional defiant disorder; SD, standard deviation; SSRI, selective serotonin reuptake inhibitor; SUD, substance use disorder.

participants with BD-I demonstrated the most severe impact on global functioning, and the highest rates of psychiatric hospitalization. Because we did not query the cause of psychiatric hospitalizations, we could not examine for differences related to whether hospitalizations were owing to manic, depressive, or other symptoms.

The ordering of groups based on highest lifetime manic symptom severity, highest in BD-I, lowest in BD-NOS, and intermediate in BD-II, aligns with the diagnostic criteria of these different BD subtypes (American Psychiatric Association 2013). Adolescents with BD-II had higher current and past depression symptoms than those with BD-I; in addition, adolescents with BD-NOS had higher current depression symptoms than those with BD-I. In the COBY study, there were no significant differences in current depressive or manic symptoms at intake between subtypes (Axelson et al. 2006). The reasons for this discrepancy are unclear and may relate to between-study differences in age (younger in COBY), recruitment approach (current sample is entirely clinically referred and treatment-seeking at enrollment), or other factors. These between-study differences may also contribute to differences in age of onset: in this study, participants with BD-NOS had the youngest age of onset, whereas in the COBY study, age of onset was significantly higher in BD-II than in BD-I and BD-NOS (Axelson et al. 2006). Implications for this finding are discussed in greater detail hereunder.

Other than the lower rates of comorbid anxiety disorders among participants with BD-I than those with BD-II and BD-NOS, there were no significant between-group differences in other comorbidities. This converges with previous findings in youth and adults in clinical and epidemiologic samples (Lewinsohn et al. 1995; McElroy et al. 2001; Birmaher and Axelson 2006; Goldstein et al. 2008; Merikangas et al. 2011).

In terms of dimensional traits, participants with BD-I has significantly lower mood lability and emotional dysregulation/borderline personality spectrum symptoms compared with those with BD-II and BD-NOS, whereas participants with BD-II and BD-NOS did not significantly differ from one another on these measures. Similarly, adults with BD-II have been found to experience greater mood lability compared with those with BD-I (Akiskal et al. 2006). These dimensional traits likely contribute in part to the observed differences in self-injurious behaviors, which were less common in BD-I than the other two groups.

Participants with BD-I were more likely than the other groups to have been treated with lithium and/or SGAs, which are first-line treatments for acute mania and for maintenance in adults and youth (Goldstein et al. 2017; Yatham et al. 2018). Of interest, rates of psychosocial treatment were highest in the BD-NOS group, followed by BD-II and finally BD-I. As we did not collect information about psychosocial treatment targets (e.g., depression, anxiety, and psychoeducation), it is unclear what factors may explain these differences. We speculate that greater emotional dysregulation and self-injurious behavior may explain in part why psychosocial treatment was more common in BD-II and BD-NOS. In terms of why psychosocial treatment was highest in BD-NOS, we speculate that nonpharmacological approaches were prioritized as a first-line approach. This may be owing to the fact of the lesser evidence base for pharmacological treatment in BD-NOS. Studies that evaluate reasons for treatment seeking are warranted to better understand these subtype-related differences.

Across all familial psychiatric disorders examined, the only between-group differences were higher rates of familial anxiety and depression in adolescents with BD-II and BD-NOS compared

with those with BD-I. Previous research from the COBY study found no differences in family psychiatric history across BD subtypes, with the exception of higher rates of familial suicide attempts among youth with BD-II (Axelson et al. 2006). The lack of subtype-related differences in family history of BD converges with findings in adults and youth (Axelson et al. 2006; Bader and Dunner 2007; Wozniak et al. 2017).

The younger age of onset, greater mood lability, and lesser use of most medication classes among adolescents with BD-NOS warrants consideration. Given that youth with BD-NOS commonly convert to another subtype of BD with time, particularly BD-I (Birmaher et al. 2006; Martinez and Fristad 2013; Towbin et al. 2013), and given that BD-NOS is associated with both younger age and greater mood lability, this raises the question of whether strategies that target reduction of mood lability among young individuals with BD may prevent or at least delay diagnostic conversion. Further research is necessary to determine the optimal treatment of BD-NOS, both in terms of reducing current symptoms and in terms of reducing the risk of future diagnostic conversion (Goldstein et al. 2017). Similar psychosocial treatment approaches have been used across different BD subtypes, including in the context of familial BD (Miklowitz et al. 2008; Goldstein et al. 2015; Vallarino et al. 2015; Miklowitz 2016). However, although there have been numerous randomized controlled trials for adolescents with BD-I, pharmacological treatment approaches for BD-II and BD-NOS are informed by much less clinical trial data than BD-I (Goldstein et al. 2017; Zalpuri and Singh 2017; Yatham et al. 2018). Although there have been some small open-label studies targeting mood symptoms in offspring of parents with BD, there have not been randomized controlled pharmacotherapy trials in this population or in BD-NOS in the absence of familial BD. Most randomized controlled trials focus on acute mania, and yet acute mania accounts for a small proportion of time relative to depression and subthreshold manic and depressive symptoms. Given that the treatment of bipolar depression and subthreshold manic symptoms is relevant across all BD subtypes, this underscores the importance of including youth across the BD spectrum within related pharmacological treatment studies.

Finally, this study replicates many of the findings found in the COBY sample in an entirely separate sample of adolescents and extends the examination of this topic to a Canadian population with universal access to a national health care system. This is a strong, effective validation of the COBY definition of BD-NOS (in comparison with other subtypes of pediatric BD) in a different and separate population. Moreover, the study focuses specifically on the mid-to-late adolescent age group that is known to be an epoch in which the prevalence of BD increases markedly (Merikangas et al. 2010; Goldstein et al. 2017).

Limitations

Several limitations warrant consideration when interpreting the present findings. First, the study included participants from a single study site based in a tertiary setting; as such, present findings may not extend to other clinical and/or community settings. Concerns about this potential limitation are mitigated by the fact that many findings from this study replicate findings from previous studies based on different samples. Second, the study is cross-sectional and retrospective, and the temporal associations between several of the study variables could not be determined. Third, despite the relatively large sample size, this study was not adequately powered to detect differences with small effect sizes. Finally, because of the

young age of the sample, a substantial proportion of youth with BD-NOS will likely go on to develop BD-I or BD-II, which may explain, in part, the limited group differences (Axelson et al. 2011).

Conclusions

Overall, this study suggests that similarities exceed differences across BD subtypes among youth and confirms multiple previous findings in this regard. One third of this study sample had a diagnosis of BD-NOS, and these adolescents demonstrated generally comparable symptoms, comorbidity, and functional impairment to adolescents with the more typical BD-I and BD-II subtypes.

Clinical Significance

This study supports the validity of BD-NOS as operationalized by the COBY study, and identifying the areas in which the different subtypes of BD converge and differ.

Disclosures

No competing financial interests exist.

References

- Akiskal HS, Kilzieh N, Maser JD, Clayton PJ, Schettler PJ, Traci Shea M, Endicott J, Scheftner W, Hirschfeld RM, Keller MB: The distinct temperamental profiles of bipolar I, bipolar II and unipolar patients. *J Affect Disord* 92:19–33, 2006.
- American Psychiatric Association: Diagnostic and Statistical Manual of Mental Disorders (5th ed.). Washington (DC), American Psychiatric Publ, 2013.
- Axelson D, Birmaher BJ, Brent D, Wassick S, Hoover C, Bridge J, Ryan N: A preliminary study of the kiddie schedule for affective disorders and schizophrenia for school-age children mania rating scale for children and adolescents. *J Child Adolesc Psychopharmacol* 13:463–470, 2003.
- Axelson D, Birmaher B, Strober M, Gill MK, Valeri S, Chiappetta L, Ryan N, Leonard H, Hunt J, Iyengar S, Bridge J, Keller M: Phenomenology of children and adolescents with bipolar spectrum disorders. *Arch Gen Psychiatry* 63:1139–1148, 2006.
- Axelson D, Birmaher B, Strober MA, Goldstein BI, Ha W, Gill MK, Goldstein TR, Yen S, Hower H, Hunt J, Liao F, Iyengar S, Dickstein D, Kim E, Ryan ND, Frankel E, Keller MB: Course of subthreshold bipolar disorder in youth: Diagnostic progression from bipolar disorder not otherwise specified. *J Am Acad Child Adolesc Psychiatry* 50:1001–1016, 2011.
- Bader CD, Dunner DL: Bipolar disorder not otherwise specified in relation to the bipolar spectrum. *Bipolar Disord* 9:860–867, 2007.
- Birmaher B, Axelson D: Course and outcome of bipolar spectrum disorder in children and adolescents: A review of the existing literature. *Dev Psychopathol* 18:1023–1035, 2006.
- Birmaher B, Axelson D, Goldstein B, Strober M, Gill MK, Hunt J, Houck P, Ha W, Iyengar S, Kim E, Yen S, Hower H, Esposito-Smythers C, Goldstein T, Ryan N, Keller M: Four-year longitudinal course of children and adolescents with bipolar spectrum disorders: The course and outcome of bipolar youth (COBY) study. *Am J Psychiatry* 166:795–804, 2009.
- Birmaher B, Axelson D, Strober M, Gill MK, Valeri S, Chiappetta L, Ryan N, Leonard H, Hunt J, Iyengar S, Keller M: Clinical course of children and adolescents with bipolar spectrum disorders. *Arch Gen Psychiatry* 63:175–183, 2006.
- Freeman AJ, Youngstrom EA, Michalak E, Siegel R, Meyers OI, Findling RL: Quality of life in pediatric bipolar disorder. *Pediatrics* 123:e446–e452, 2009.
- Fristad MA, Algorta GP: Future directions for research on youth with bipolar spectrum disorders. *J Clin Child Adolesc Psychol* 42:734–747, 2013.
- Gerson AC, Gerring JP, Freund L, Joshi P, Capozzoli J, Brady K, Denckla MB: The children's affective lability scale: A psychometric evaluation of reliability. *Psychiatry Res* 65:189–197, 1996.
- Goldstein BI, Birmaher B: Prevalence, clinical presentation and differential diagnosis of pediatric bipolar disorder. *Isr J Psychiatry Relat Sci* 49:3–14, 2012.
- Goldstein BI, Birmaher B, Carlson GA, DelBello MP, Findling RL, Fristad M, Kowatch RA, Miklowitz DJ, Nery FG, Perez-Algorta G, Van Meter A, Zeni CP, Correll CU, Kim HW, Wozniak J, Chang KD, Hillegers M, Youngstrom EA: The International Society for Bipolar Disorders Task Force report on pediatric bipolar disorder: Knowledge to date and directions for future research. *Bipolar Disord* 19:524–543, 2017.
- Goldstein BI, Strober M, Axelson D, Goldstein TR, Gill MK, Hower H, Dickstein D, Hunt J, Yen S, Kim E, Ha W, Liao F, Fan J, Iyengar S, Ryan ND, Keller MB, Birmaher B: Predictors of first-onset substance use disorders during the prospective course of bipolar spectrum disorders in adolescents. *J Am Acad Child Adolesc Psychiatry* 52:1026–1037, 2013.
- Goldstein BI, Strober MA, Birmaher B, Axelson DA, Esposito-Smythers C, Goldstein TR, Leonard H, Hunt J, Gill MK, Iyengar S, Grimm C, Yang M, Ryan ND, Keller MB: Substance use disorders among adolescents with bipolar spectrum disorders. *Bipolar Disord* 10:469–478, 2008.
- Goldstein TR, Birmaher B, Axelson D, Ryan ND, Strober MA, Gill MK, Valeri S, Chiappetta L, Leonard H, Hunt J, Bridge JA, Brent DA, Keller M: History of suicide attempts in pediatric bipolar disorder: Factors associated with increased risk. *Bipolar Disord* 7: 525–535, 2005.
- Goldstein TR, Fersch-Podrat RK, Rivera M, Axelson DA, Merranko J, Yu H, Brent DA, Birmaher B: Dialectical behavior therapy for adolescents with bipolar disorder: Results from a pilot randomized trial. *J Child Adolesc Psychopharmacol* 25:140–149, 2015.
- Hollingshead A: Four factor index of social status. *Yale J Sociol* 8:21–52, 1975.
- IBM Corp: IBM SPSS statistics for windows, Version 24.0. Armonk (NY), IBM Corp., 2016.
- Judd LL, Akiskal HS, Schettler PJ, Coryell W, Maser J, Rice JA, Solomon DA, Keller MB: The comparative clinical phenotype and long-term longitudinal episode course of bipolar I and II: A clinical spectrum or distinct disorders? *J Affect Disord* 73:19–32, 2003.
- Judd LL, Schettler PJ, Solomon DA, Maser JD, Coryell W, Endicott J, Askisal HS: Psychosocial disability and work role function compared across the long-term course of bipolar I, bipolar II and unipolar major depressive disorders. *J Affect Disord* 108:49–58, 2008.
- Kaufman J, Birmaher B, Brent D, Rao U, Flynn C, Moreci P, Williamson D, Ryan N: Schedule for affective disorders and schizophrenia for school-age children-present and lifetime version (K-SADS-PL): Initial reliability and validity data. *J Am Acad Child Adolesc Psychiatry* 36:980–988, 1997.
- Lewinsohn PM, Klein DN, Seeley JR: Bipolar disorders in a community sample of older adolescents: Prevalence, phenomenology, comorbidity, and course. *J Am Acad Child Adolesc Psychiatry* 34: 454–463, 1995.
- Martinez MS, Fristad MA: Conversion from bipolar disorder not otherwise specified (BP-NOS) to bipolar I or II in youth with family history as a predictor of conversion. *J Affect Disord* 148:431–434, 2013.

- Masi G, Perugi G, Millepiedi S, Mucci M, Pari C, Pfanner C, Berloff S, Toni C: Clinical implications of DSM-IV subtyping of bipolar disorders in referred children and adolescents. *J Am Acad Child Adolesc Psychiatry* 46:1299–1306, 2007.
- McElroy SL, Altshuler LL, Suppes T, Keck PE, Frye MA, Denicoff KD, Nolen WA, Kupka RW, Leverich GS, Rochussen JR, Rush AJ, Post RM: Axis I psychiatric comorbidity and its relationship to historical illness variables in 288 patients with bipolar disorder. *Am J Psychiatry* 158:420–426, 2001.
- Merikangas KR, He JP, Burstein M, Swanson SA, Avenevoli S, Cui L, Benjet C, Georgiades K, Swendsen J: Lifetime prevalence of mental disorders in U.S. adolescents: Results from the National Comorbidity Survey Replication—Adolescent Supplement (NCS-A). *J Am Acad Child Adolesc Psychiatry* 49:980–989, 2010.
- Merikangas KR, Jin R, He JP, Kessler RC, Lee S, Sampson NA, Viana MC, Andrade LH, Hu C, Karam EG, Ladea M, Medina-Mora ME, Ono Y, Posada-Villa J, Sagar R, Wells JE, Zarkov Z: Prevalence and correlates of bipolar spectrum disorder in the world mental health survey initiative. *Arch Gen Psychiatry* 68:241–251, 2011.
- Miklowitz DJ: Evidence-based family interventions for adolescents and young adults with bipolar disorder. *J Clin Psychiatry* 77(Suppl E1):e5, 2016.
- Miklowitz DJ, Axelson DA, Birmaher B, George EL, Taylor DO, Schneck CD, Beresford CA, Dickinson LM, Craighead WE, Brent DA: Family-focused treatment for adolescents with bipolar disorder. *Arch Gen Psychiatry* 65:1053, 2008.
- Parker G, Fletcher K, McCraw S, Futeran S, Hong M: Identifying antecedent and illness course variables differentiating bipolar i, bipolar ii and unipolar disorders. *J Affect Disord* 148:202–209, 2013.
- Rathus JH, Wagner D, Miller AL: Psychometric evaluation of the life problems inventory, a measure of borderline personality features in adolescents. *J Psychol Psychother* 5:198, 2015.
- Rizzo CJ, Esposito-Smythers C, Swenson L, Birmaher B, Ryan N, Strober M, Chiappetta L, Valeri S, Hunt J, Axelson D, Leonard H, Keller M: Factors associated with mental health service utilization among bipolar youth. *Bipolar Disord* 9:839–850, 2007.
- Shaffer D, Gould MS, Brasic J, Ambrosini P, Fisher P, Bird H, Aluwahlia S: A Children's Global Assessment Scale (CGAS). *Arch Gen Psychiatry* 40:1228–1231, 1983.
- Towbin K, Axelson D, Leibenluft E, Birmaher B: Differentiating bipolar disorder—not otherwise specified and severe mood dysregulation. *J Am Acad Child Adolesc Psychiatry* 52:466–481, 2013.
- Vallarino M, Henry C, Etain B, Gehue LJ, Macneil C, Scott EM, Barbato A, Conus P, Hlastala SA, Fristad M, Miklowitz DJ, Scott J: An evidence map of psychosocial interventions for the earliest stages of bipolar disorder. *Lancet Psychiatry* 2:548–563, 2015.
- Van Meter AR, Burke C, Kowatch RA, Findling RL, Youngstrom EA: Ten-year updated meta-analysis of the clinical characteristics of pediatric mania and hypomania. *Bipolar Disord* 18:19–32, 2016.
- Weissman MM, Wickramaratne P, Adams P, Wolk S, Verdelli H, Olfson M: Brief screening for family psychiatric history: The family history screen. *Arch Gen Psychiatry* 57:675–682, 2000.
- Wozniak J, Uchida M, Faraone SV, Fitzgerald M, Vaudreuil C, Carrellas N, Davis J, Wolenski R, Biederman J: Similar familial underpinnings for full and subsyndromal pediatric bipolar disorder: A familial risk analysis. *Bipolar Disord* 19:168–175, 2017.
- Yatham LN, Kennedy SH, Parikh SV, Schaffer A, Bond DJ, Frey BN, Sharma V, Goldstein BI, Rej S, Beaulieu S, Alda M, MacQueen G, Milev RV, Ravindran A, O'Donovan C, McIntosh D, Lam RW, Vazquez G, Kapczinski F, McIntyre RS, Kozicky J, Kanba S, Lafer B, Suppes T, Calabrese JR, Vieta E, Malhi G, Post RM, Berk M: Canadian Network for Mood and Anxiety Treatments (CANMAT) and International Society for Bipolar Disorders (ISBD) 2018 guidelines for the management of patients with bipolar disorder. *Bipolar Disord* 20:97–170, 2018.
- Zalpur I, Singh MK: Treatment of psychiatric symptoms among offspring of parents with bipolar disorder. *Curr Treat Options Psychiatry* 4:341–356, 2017.

Address correspondence to:
Benjamin I. Goldstein, MD, PhD
Department of Psychiatry
Centre for Youth Bipolar Disorder
Sunnybrook Health Sciences Centre
2075 Bayview Avenue
Toronto M4N 3M5
Ontario
Canada

E-mail: benjamin.goldstein@sunnybrook.ca